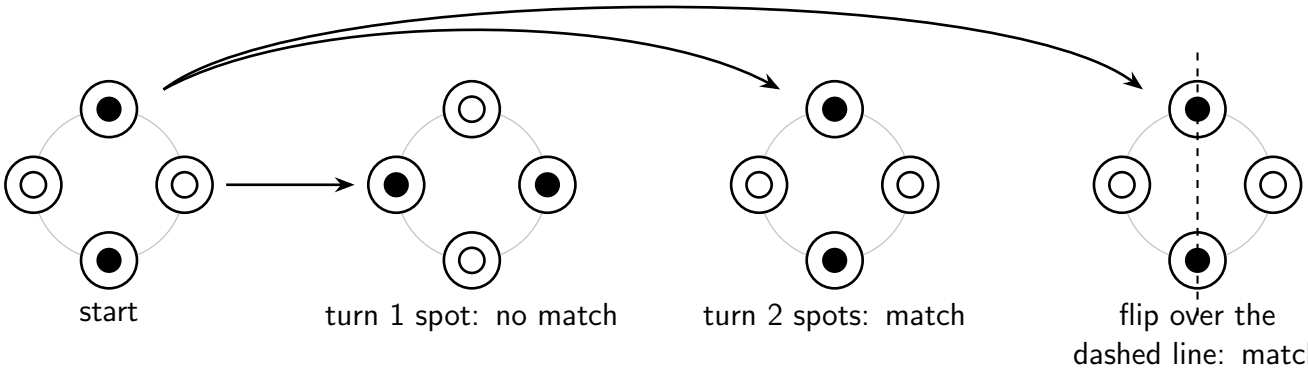
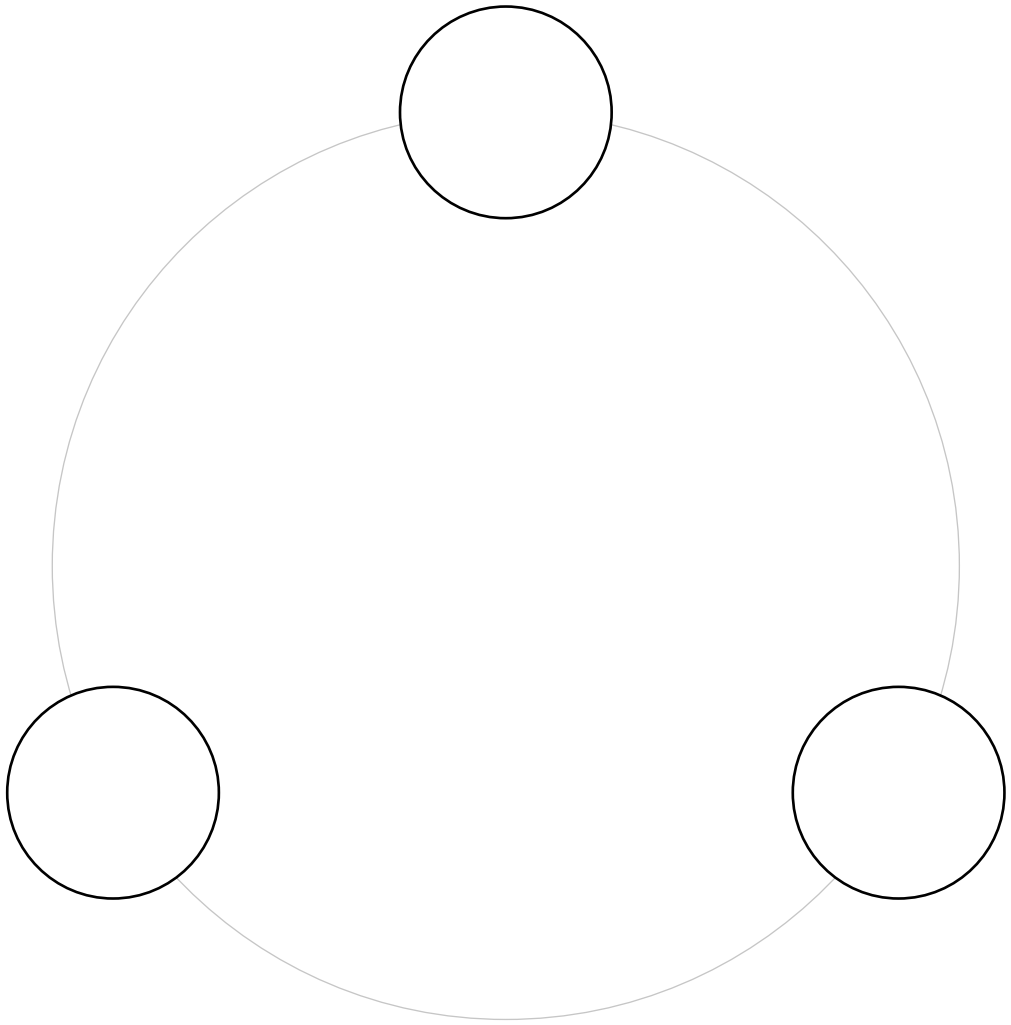
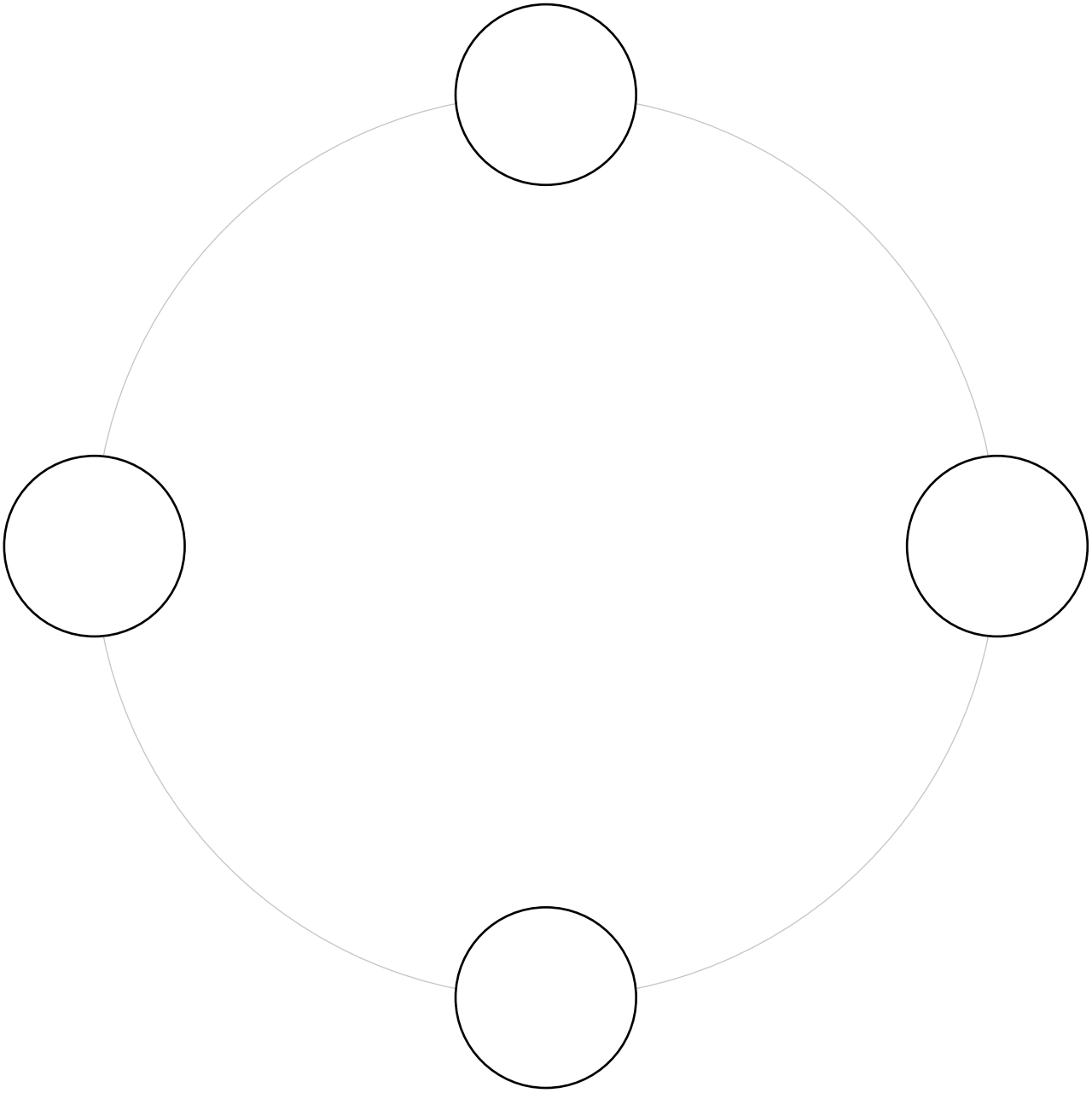
Fill every spot; neighbors may be the same kind. Only counter kinds and spots count. Keep the original still. Turn or flip (turn over) its tracing copy until all spots line up. A whole turn does not count.



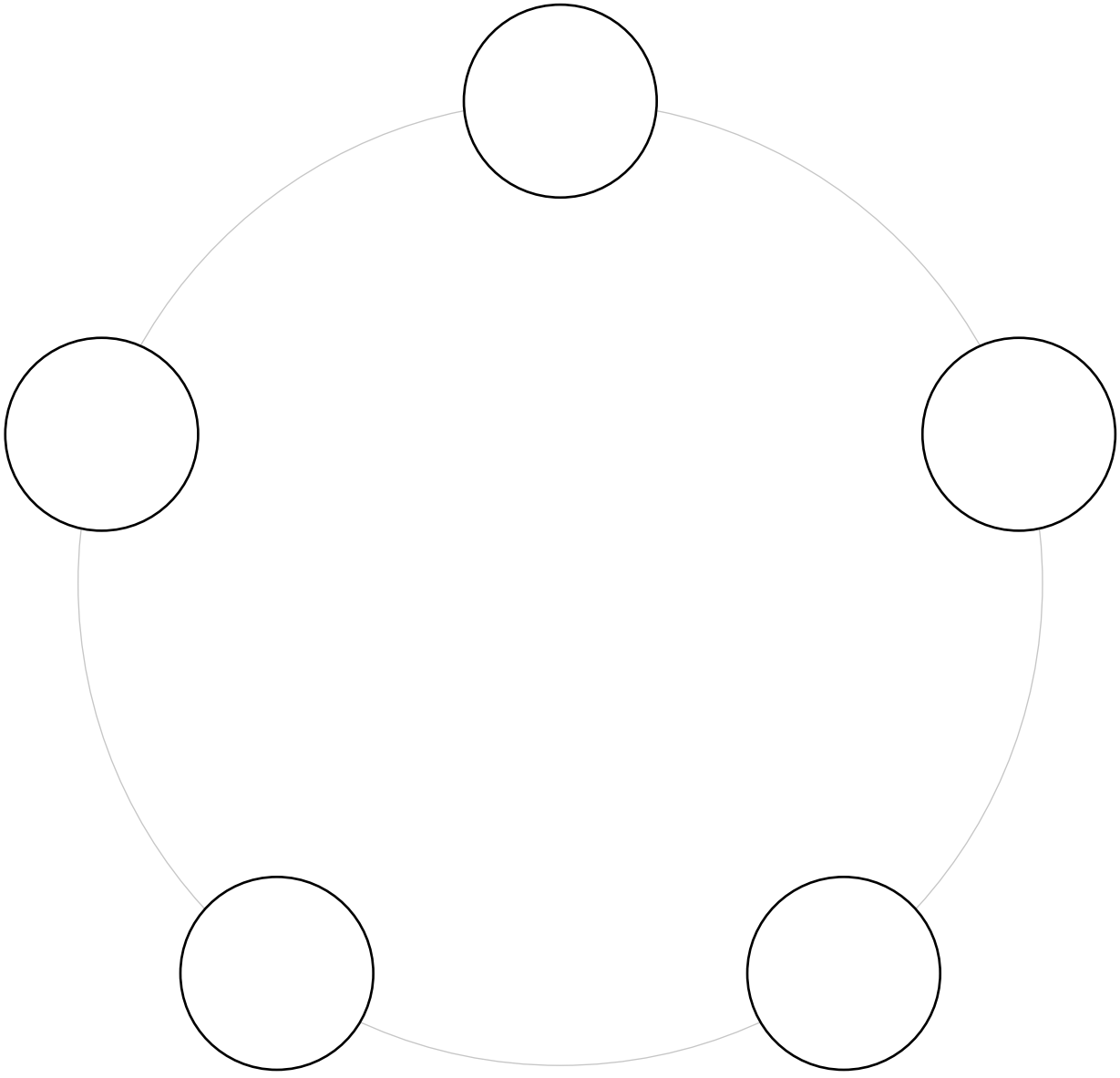
Problem 1: Find the fewest kinds of counter for a pattern on the three-ring that no turn or flip can match. Can you rule out every pattern with fewer kinds?



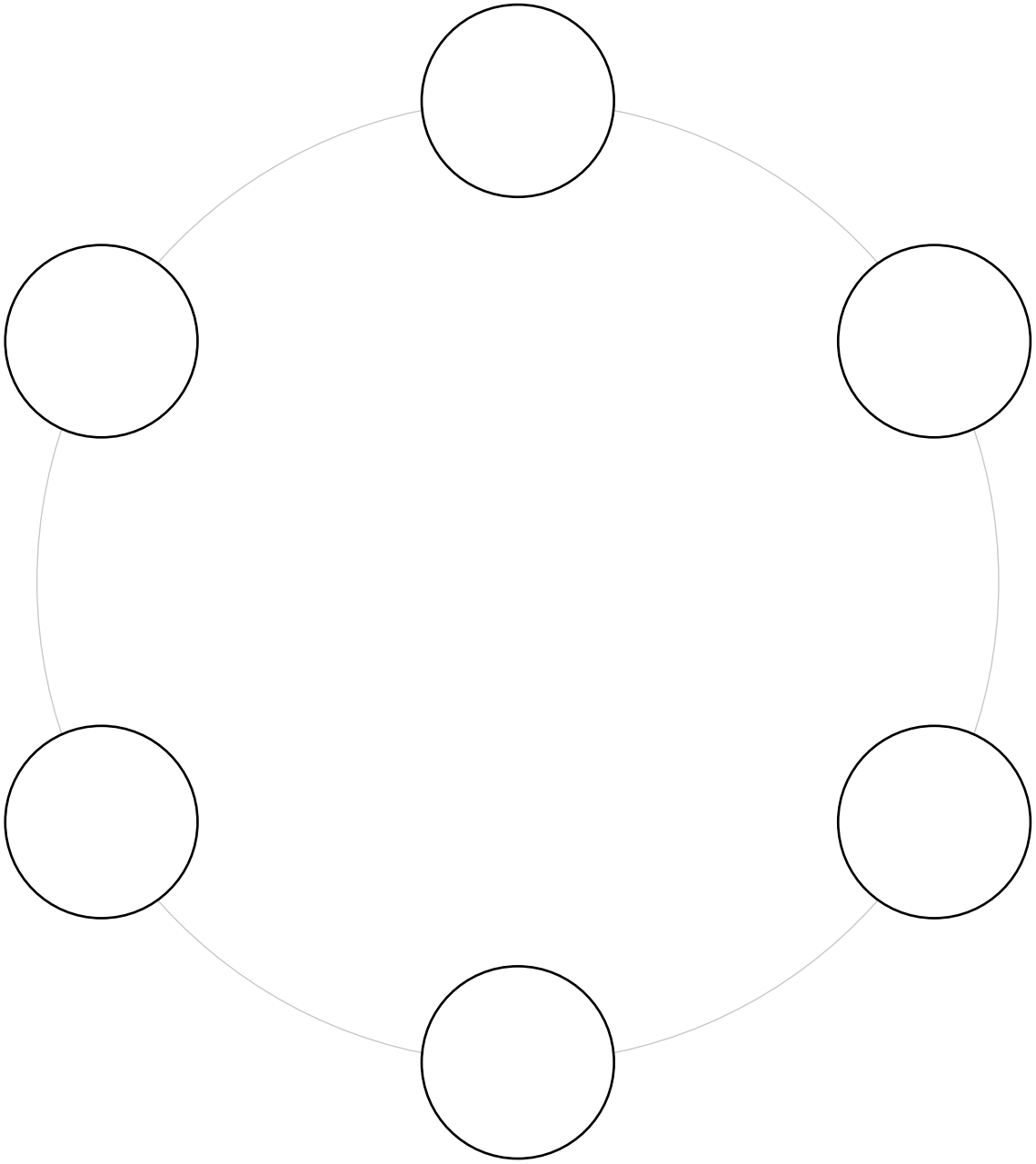
Problem 2: Find the fewest kinds of counter for a pattern on the four-ring that no turn or flip can match. Explain why fewer kinds cannot work.



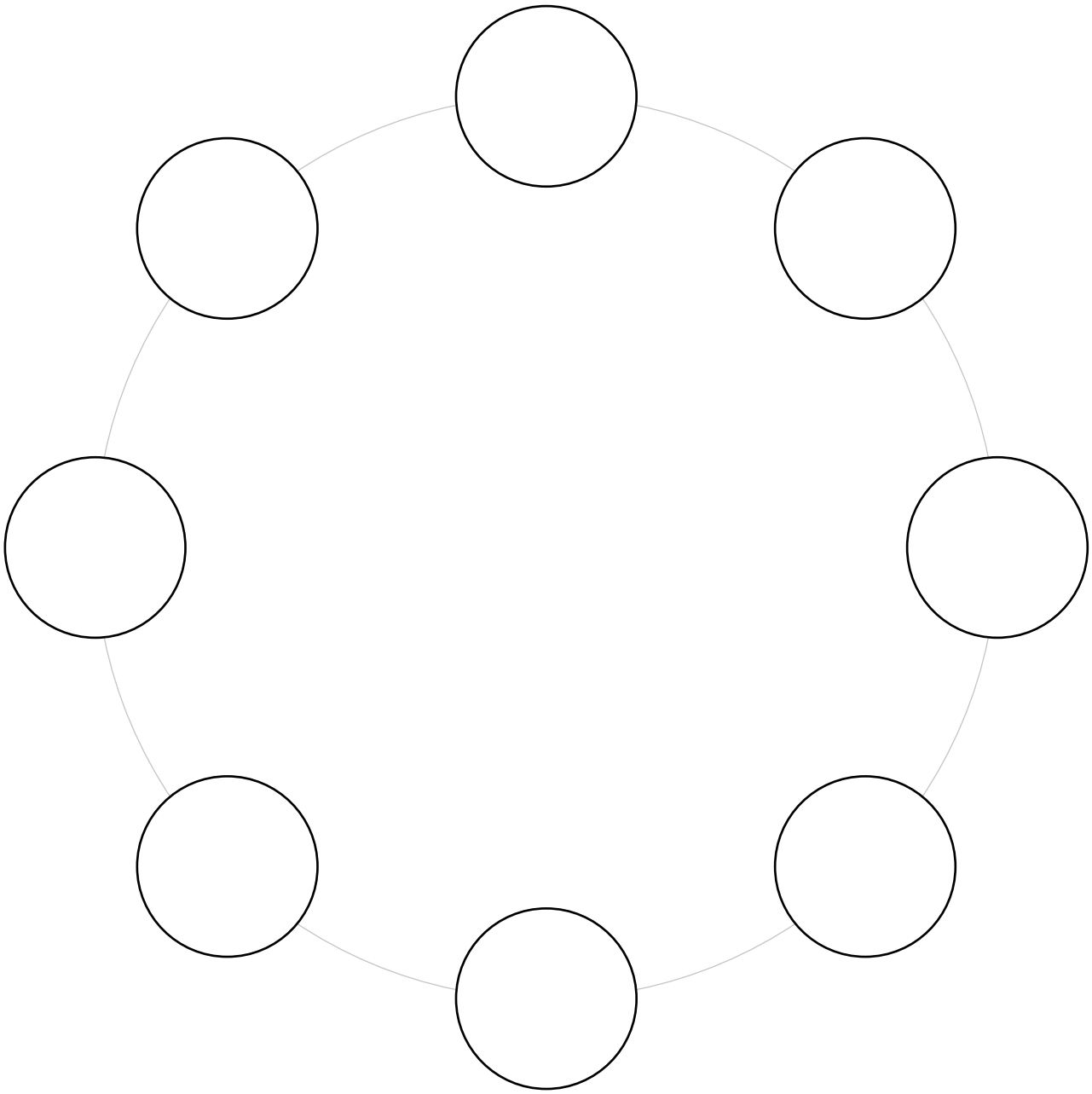
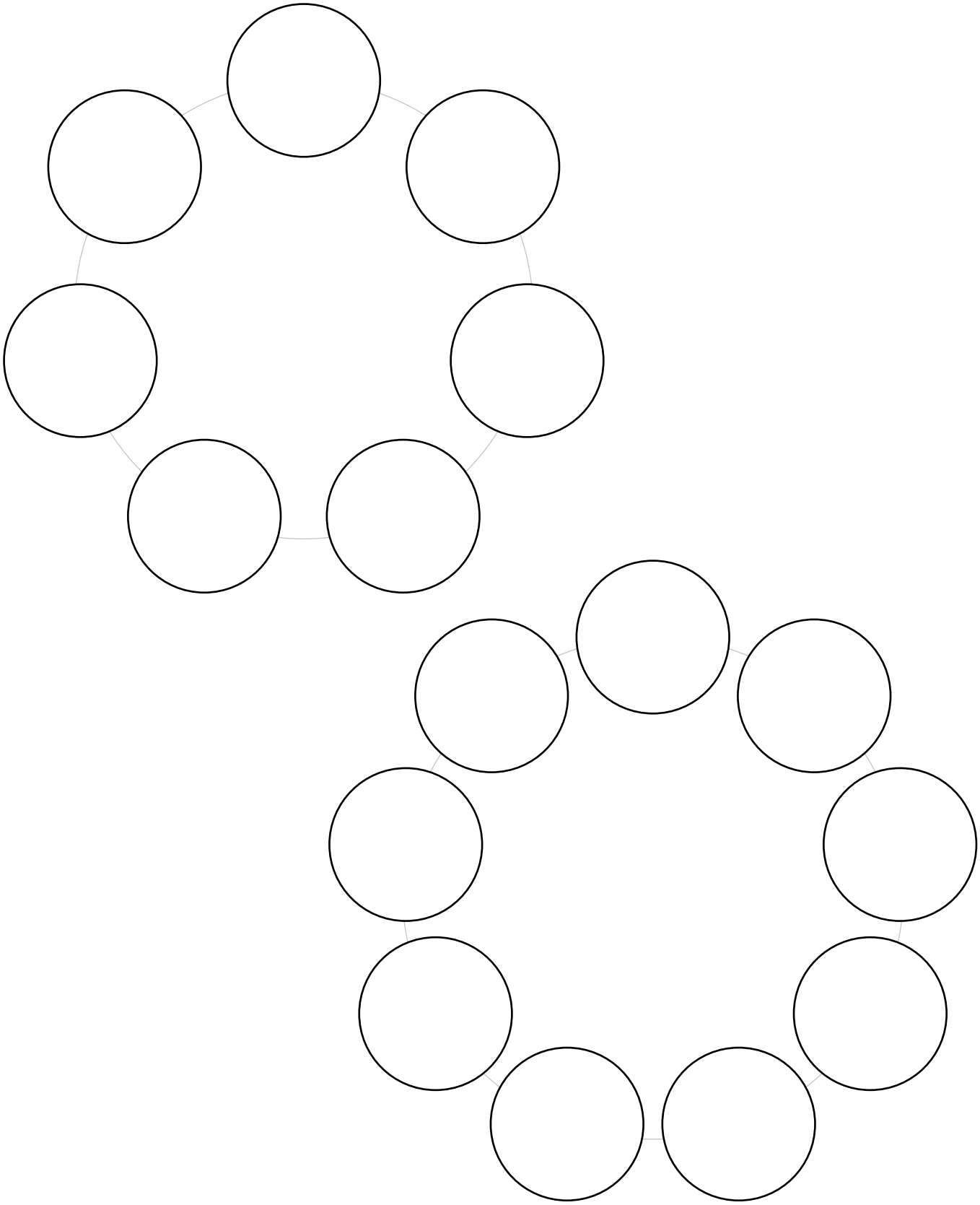
Problem 3: Find the fewest kinds of counter for a pattern on the five-ring that no turn or flip can match. Find an explanation that also works for the three-ring and four-ring.



Problem 4: Use two kinds of counter to make a pattern on this six-ring that no turn or flip can match. Explain why each turn and flip fails.



Problem 5: Can you make a two-kind pattern that no turn or flip can match on any ring with at least six spots? Test your idea on this eight-ring, then explain why it always works.



Extra workspace for Problem 5: a seven-ring and a nine-ring.